

Muhammad Ekbal Ahmad

Software Engineer & Researcher

Hamburg, Germany | it2auto@gmail.com | muhammadekbal.github.io
muhammad-ekbal-ahmad-18913b99 | muhammadekbal | 0009-0003-5258-9291
Muhammad-Ekbal-Ahmad | Google Scholar

Education

Carl von Ossietzky University Oldenburg, Computer Science — Oldenburg, Germany
- Master's thesis — Jan 2017 – Jan 2020

Tishreen University, Computer Science and Automatic Control — Syria
Jan 2003 – Jan 2008

Experience

TUHH-IDAC, Scientific Researcher — Hamburg, Germany
2021 – present
5 years

OFFIS, Scientific Researcher (WIMI) — Oldenburg, Germany
Research work associated with the ViVre project.
Oct 2020 – 2021
1 year

OFFIS, Research Assistant (WIHI) — Oldenburg, Germany
Research work associated with the Designetz project.
Nov 2018 – Oct 2020
2 years
- Frontend development using Angular.
- Data visualization using D3.js.
- Graph visualization using Graphviz.
- Geographic visualization using Leaflet.

Project Group RIO, Frontend Developer — Oldenburg, Germany
Development of a navigation application.
Oct 2018 – Sept 2019
1 year
- Frontend development using Angular.
- Mobile application development using Apache Cordova.

OFFIS, Research Assistant (WIHI) — Oldenburg, Germany
Research work associated with the IKIMUNI project.
Nov 2017 – Oct 2018
1 year
- User interface development using Qt Quick and QML.
- Software development using C++.
- Industrial communication using OPC UA.

Publications

Workshops

Staying Calm and Full of Energy — Stress Competence for PhD Candidates — Nov 2021

Introduction to the Rules of Good Scientific Practice — Nov 2021

Perfekt ins Job einsteigen — Mar 2021

Skills

Programming Languages

Frontend Development

Visualization and Mapping

Mobile Development

UI Development

Communication Protocols

Research Interests

Languages

German | English | Arabic | Kurdish

Interests

Research

Applications

Certificates

Projects

Multi-Robot Collaboration for Monitoring and Inspection of Civil Engineering Structures Oct 2026 – present
Research project on multi-robot collaboration for monitoring and inspecting civil engineering structures.
- Project details: https://www.tuhh.de/idac/research/multi-robot-collaboration

Modeling and Design Concepts for Digital Roads Oct 2023 – Oct 2026
Research project focused on modeling and design concepts for digital road infrastructure.
- Project details: https://www.tuhh.de/idac/research/digital-roads

Semi-Probabilistic, Sensor-Based Design Concepts for Intelligent Structural Systems Oct 2021 – May 2022
Research project investigating sensor-based design concepts for intelligent structural systems.
- Project details: https://www.tuhh.de/idac/research/intelligent-structures

ViVre Oct 2020 – 2021
Research project associated with scientific work at OFFIS.
- Scientific research at OFFIS.

Designetz Nov 2018 – Oct 2020
Research project involving frontend development and visualization.
- Angular frontend development.
- Data visualization with D3.js.
- Graph visualization with Graphviz.
- Geographic visualization with Leaflet.

Navigation Application Oct 2018 – Sept 2019

Navigation application developed with Project Group RIO.

- Angular frontend.
- Mobile support using Apache Cordova.

IKIMUNI Nov 2017 – Oct 2018

Research project involving user interface and software development.

- Qt Quick and QML.
- C++ development.
- OPC UA communication.

References